

Fixed-Batch CVaR and Participation in a Quadratic Resource-Allocation Mechanism

Q A Systematic Approach to Mechanism Design with Stochastic Dynamic Stability [arXiv:2608.29130](#)

P Distributed risk-averse optimization via CVaR [arXiv:2609.04460](#)

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Abstract. We examine a risk-adjusted use of the quadratic resource-allocation mechanism of Garjani, Shokri and Kebriaei with the empirical conditional value-at-risk (CVaR) oracle studied by Wang, Huang, Xu and Johansson. A one-sample example has strongly convex true and expected empirical CVaR losses, yet the exact surrogate-game equilibrium gives each agent true-risk utility $-3/32$ relative to opting out. Under a uniform value-error bound and an exact surrogate equilibrium, we obtain a $2e_n$ bound on unilateral incentives and participation, and a $2\sum_n e_n$ bound on separable-risk welfare when the surrogate planner is implemented. These are equilibrium statements, not a convergence theorem for a strategic learning algorithm.

1 Setting and relation to prior work

Garjani, Shokri and Kebriaei (Q) construct quadratic payments for strategic agents with private expected valuations and local resource constraints. Under their assumptions, the induced game has a unique Nash equilibrium that implements the expected-welfare optimum with budget balance and individual rationality; their variable-sample proximal best-response method converges in mean square. Wang, Huang, Xu and Johansson (P) instead study cooperative minimisation of local CVaRs at a shared decision. Their projected, one-point method uses empirical CVaR, and its fixed-batch result concerns an optimum of an expected empirical surrogate. It does not establish convergence in Q’s strategic game [2, 3].

Two parts of the surrounding argument are already in the literature. Wang, Shen, Zavlanos and Johansson give an example in which a strongly monotone risk-neutral game loses uniqueness under CVaR, and analyse learning in strongly monotone risk-averse games [4]. Cherukuri proves consistency and finite-sample guarantees for sample-average CVaR variational inequalities, including a routing equilibrium application [1]. Thus loss of curvature under a risk change and sensitivity of CVaR equilibria to sampling are published steps. Our claim is the narrower, mechanism-specific fixed-batch participation failure and the conditional value-level consequences recorded below; we make no priority claim for either ingredient.

Let $J_n(x_n, \xi_n) = -\psi_n(x_n, \xi_n)$ be an agent’s loss and let $\beta_n \in (0, 1)$ denote upper-tail mass. Write

$$C_n(x_n) = \text{CVaR}_{\beta_n}[J_n(x_n, \xi_n)], \quad R_n(x_n) = -C_n(x_n).$$

At a message profile s , Q’s deterministic tax $t_n(s)$ gives risk-adjusted utility $R_n(x_n) - t_n(s)$ by CVaR cash translation. The associated welfare target is $\sum_n R_n(x_n)$, with the same capacity and local constraints as in Q. Participation is measured against $R_n(0)$, which may be normalised to zero.

Q assumes strong concavity of the *expected* valuation. This alone does not give strong concavity of R_n : on $[0, 1]$, two equiprobable losses 0 and $x^2 - 1$ have strongly convex expectation while their upper-half CVaR is identically zero. Uniform samplewise m -strong convexity is a sufficient curvature repair, since cash translation makes $C_n(x) - m\|x\|^2/2$ convex. It need not make CVaR differentiable, so applying all of Q’s smooth-game conclusions still requires suitable nonsmooth and constraint arguments. The general loss-of-uniqueness phenomenon was also exhibited in [4].

2 A fixed-batch participation failure

The example concerns the *expected empirical* CVaR objective to which a repeatedly sampled, fixed-batch oracle is directed. It makes no assertion that P’s algorithm can run on Q’s feasible set or that Q’s learning process converges to this equilibrium.

Proposition 1. *For $N \geq 2$ agents with $x_n \in [0, 1]$, capacity $c = N$, and Q 's explicit quadratic tax, there is a one-sample CVaR surrogate game with an exact equilibrium at $x_n = 1/4$ and zero price bids. Each agent's true-risk utility there is $-3/32 < R_n(0) = 0$, although both the true and surrogate CVaR losses are strongly convex.*

Proof. Take independent, equiprobable $\xi_n \in \{0, 1\}$, tail mass $\beta_n = 1/2$, and

$$J_n(x, \xi_n) = \frac{x^2}{2} - \frac{3x}{4} + \xi_n x.$$

For $x \geq 0$, the upper-half loss is the $\xi_n = 1$ loss, so

$$C_n(x) = \frac{x^2}{2} + \frac{x}{4}, \quad \tilde{C}_n(x) := \mathbb{E}_{\text{batch}} \hat{C}_n(x) = \frac{x^2}{2} - \frac{x}{4}.$$

With one observation, empirical CVaR equals that observation. The true minimum is $x = 0$; the surrogate minimum is $x = 1/4$. Both second derivatives equal one. When the other agents bid zero, Q 's explicit tax reduces to

$$t_n = \alpha \left[\frac{p_n^2}{2} + \frac{p_n}{N-1} \left(c - x_n - \sum_{m \neq n} x_m \right) \right].$$

Here $\alpha > 0$ and feasible price bids satisfy $p_n \geq 0$. Because $c = N$ and each feasible allocation lies in $[0, 1]$, this tax is nonnegative for every feasible unilateral allocation and price bid. It is zero at $p_n = 0$, where maximising surrogate utility amounts to minimising $\tilde{C}_n$. Thus $x_n = 1/4, p_n = 0$ is an exact equilibrium of the surrogate game. There $C_n(1/4) = 3/32$, whereas $C_n(0) = 0$, proving the stated true-risk participation failure. This is a failure under the true CVaR valuation, not a claim about every realised payoff. $\square$

3 Conditional value-level guarantees

Let $\tilde{R}_n$ be the negative expectation, over a fresh batch and smoothing direction, of P 's empirical CVaR at a valid perturbed point. Suppose $|J_n| \leq U_n$, local CVaR is L_n -Lipschitz, batch size is s_n , and every smoothing query used below is valid. P 's smoothing estimate and its empirical-distribution bound imply, uniformly on the domain under consideration,

$$\sup_x |\tilde{R}_n(x) - R_n(x)| \leq e_n, \quad e_n := L_n \delta_n + \frac{U_n \sqrt{2\pi}}{\beta_n \sqrt{s_n}}.$$

This is a deterministic bound for the expected empirical surrogate. It is not a uniform guarantee for a particular realised batch.

Proposition 2. *Suppose $\tilde{s}$ is an exact Nash equilibrium of Q 's tax game with valuations $\tilde{R}_n$, and the preceding bound holds for all allocations in every unilateral deviation. Then no agent can improve its true-risk utility by more than $2e_n$ through a unilateral deviation. If Q 's opt-out deviation has nonpositive tax at $\tilde{s}$, then*

$$R_n(\tilde{x}_n) - t_n(\tilde{s}) - R_n(0) \geq -2e_n.$$

If, in addition, Q 's implementation conditions ensure that $\tilde{x}$ maximises $\sum_n \tilde{R}_n$ on the feasible allocation set, then its true separable-risk welfare gap is at most $2 \sum_n e_n$. If the surrogate game also meets Q 's budget-balance conditions, with the symmetric equilibrium prices and planner complementarity used in Q 's proof, the taxes sum to zero at this equilibrium.

Proof. For any unilateral deviation, substitute R_n for $\tilde{R}_n$ on each side of the surrogate Nash inequality. The fixed tax at each message profile is unchanged, and the two valuation substitutions cost at most e_n each. Q 's opt-out message with zero allocation then yields the participation inequality when its tax is nonpositive. For welfare, insert $\sum_n \tilde{R}_n$ between true welfare at a true optimum and at $\tilde{x}$; surrogate optimality makes its middle difference nonpositive, and the two remaining differences are each bounded by $\sum_n e_n$. For balance, Q evaluates its tax at a surrogate planner allocation with symmetric prices, then uses planner complementarity and its P3 coefficient conditions. Without those implementation properties, its budget proof does not apply to an arbitrary surrogate equilibrium. $\square$

The ledger also records a conditional equilibrium-distance calculation. If a feasible smoothing oracle exists and the smoothed strategic operator is strongly monotone with constant $\nu > 0$, comparison of its two variational inequalities with P’s gradient-error bound gives

$$\|\tilde{s}_{\delta,s} - s_{\delta}\| \leq \frac{1}{\nu} \sqrt{\sum_n \frac{2\pi d_n^2 U_n^2}{\beta_n^2 \delta_n^2 s_n}}.$$

Here both equilibria are for smoothed games, with true and expected empirical CVaR respectively. This sensitivity statement assumes the required operator and oracle; it is not a learning-dynamics result.

4 Interpretation

Deterministic payments with individual CVaR valuations optimise the sum of local CVaRs, not generally CVaR of aggregate loss. To see the distinction, take two agents with $x_1, x_2 \geq 0$, $x_1 + x_2 \leq 1$, and equally likely, mutually exclusive scenarios $(Z_1, Z_2) = (1, 0)$ and $(0, 1)$. Set

$$J_1 = 2x_1 Z_1 + qx_1^2 - Bx_1, \quad J_2 = x_2 Z_2 + qx_2^2 - Bx_2, \quad 0 < q < \frac{1}{2}, \quad B > 2 + 2q,$$

and take upper-tail mass $\beta = 1/2$. Every sample loss is strongly convex. The sum of local CVaRs is $2x_1 + x_2 + q(x_1^2 + x_2^2) - B(x_1 + x_2)$, uniquely minimised at $(0, 1)$. CVaR of aggregate loss is $\max\{2x_1, x_2\} + q(x_1^2 + x_2^2) - B(x_1 + x_2)$, uniquely minimised at $(1/3, 2/3)$. The aggregate-risk regret at the local-risk solution is exactly $1/3 + 4q/9 > 0$. Scenario-contingent pooling transfers yield an algebraic risk-aggregation identity if realised losses are verifiable, but no incentive-compatible pooling mechanism is established.

5 Limitations and open questions

The main geometric obstacle is immediate: P’s perturbation construction assumes an origin-centred ball inside the feasible set, while Q’s allocations are nonnegative and its electric-vehicle example has affine flow equalities. A relative-interior oracle or a justified extension of the loss would be needed. Q’s strategic proximal best-response convergence theorem does not cover P’s cooperative consensus update or its biased CVaR estimates. The propositions here require an exact surrogate equilibrium; they do not show that agents learn one. They also leave exact true-risk participation unsupported without a positive margin exceeding $2e_n$. Establishing a feasible oracle and an inexact strategic learning theorem remains open.

References

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